CI/CD Deployment of Brain Tasks Application Using AWS EKS

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1. Introduction

This project demonstrates the implementation of a complete Continuous Integration and Continuous Deployment (CI/CD) pipeline for deploying a containerized web application on Amazon Web Services (AWS).

The solution uses the following AWS services:

- Amazon Elastic Container Registry (ECR)
- AWS CodeBuild
- Amazon Elastic Kubernetes Service (EKS)
- AWS CloudWatch
- AWS Identity and Access Management (IAM)

The application is automatically built, containerized, stored in ECR, deployed to an EKS cluster, and monitored using CloudWatch.

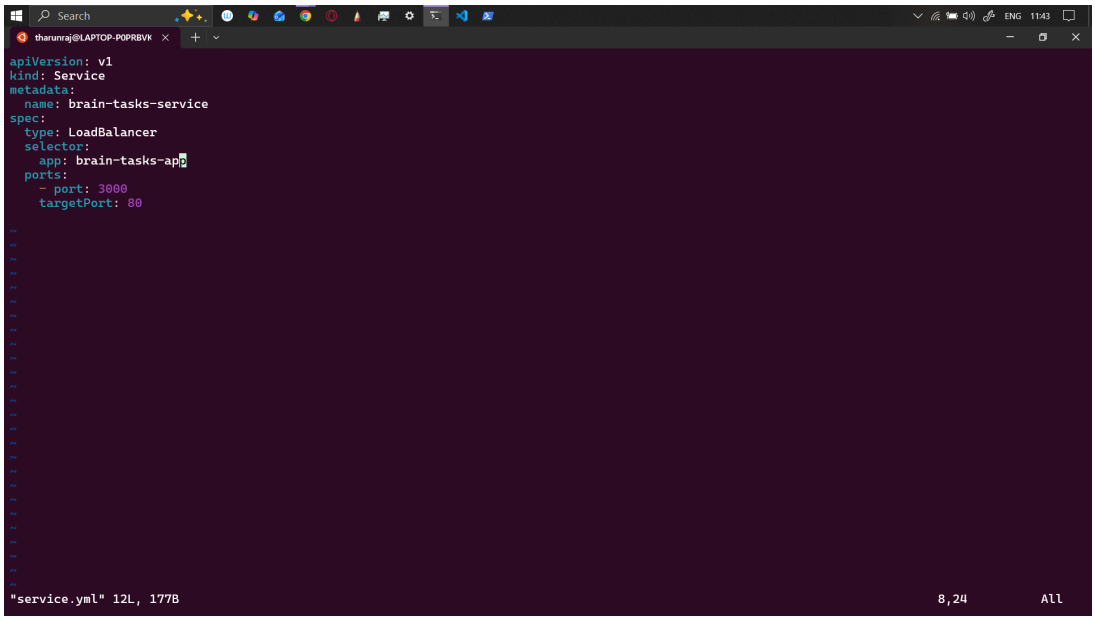
2. Project Objective

The objective of this project is to:

- Containerize a web application using Docker
- Automate the build and deployment process
- Deploy the application to a managed Kubernetes cluster (EKS)
- Expose the application publicly via LoadBalancer
- Implement centralized logging and monitoring using CloudWatch

2. Image creation

- Clone the given repository to your local machine using **“git clone <repo-name>”**. The given repository has all the prerequisites for this app.
- Within the root repository create a Dockerfile to create the image.
- After creating the Dockerfile you can containerize the application using the command **"docker build -t <app-name>”** .
- Test the app using the command **“docker run -p 3000:80 <app-name>”**.
- You can push the image by tagging it to your repository first by using the command **“docker tag docker-user/docker-repo-name:tag-name”**.
- Push the image to dockerhub by using **“docker push docker-user/docker-repo-name:tag-name”**.



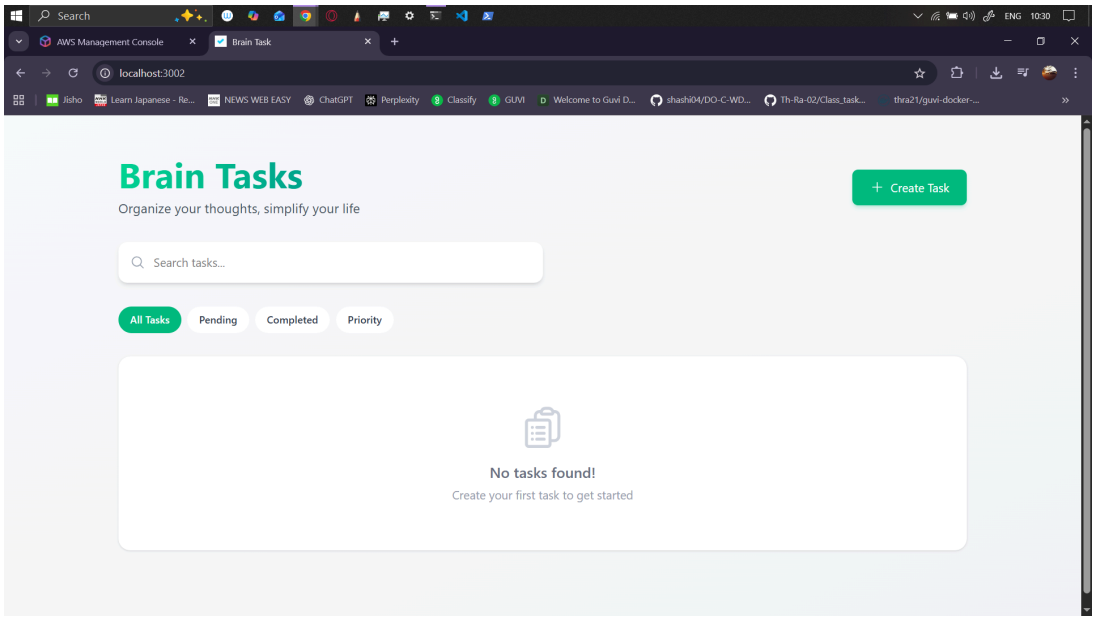


Fig 1.1 and 1.2

3. Architecture Overview

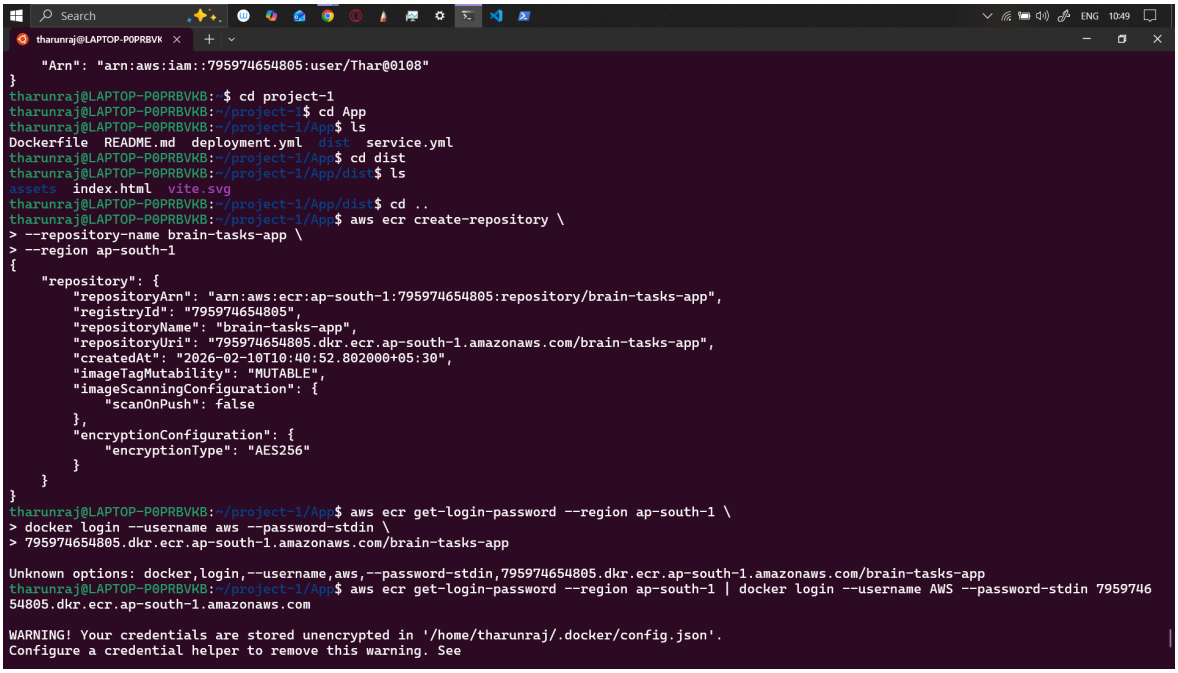
Deployment Flow:

1. The developer pushes source code.
2. AWS CodeBuild is triggered.
3. Docker image is built.
4. Image is pushed to Amazon ECR.
5. Kubernetes manifests are applied to EKS.
6. EKS pulls the image from ECR.
7. Application is exposed using a LoadBalancer.
8. CloudWatch collects build logs, application logs, and cluster metrics.

4. Infrastructure Components

4.1 Amazon ECR

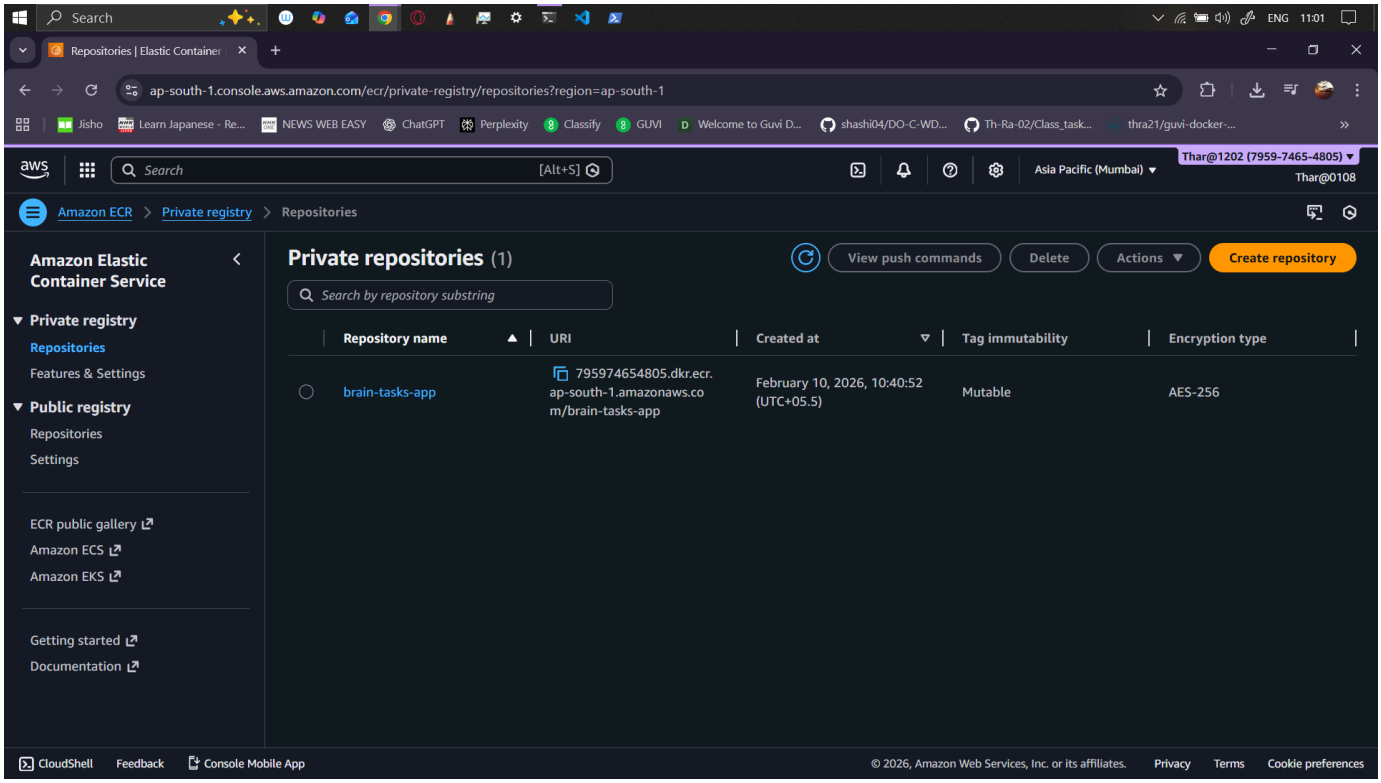
- Create an AWS ECR repository using the following command -
- "**aws ecr create-repository **"
- "**--repository-name brain-tasks-app **"
- "**--region ap-south-1**"
- Note the repo uri
- Connect the repo to your local machine using the command -
- "**aws ecr get-login-password –region ap-south-1 | docker login –username AWS –password-stdin <repo-uri>**"
- Now you can tag and push the image using the same way you do it in dockerhub.
- Check your created repo and image in your aws console.
- This provides secure image storage for Kubernetes deployment.



```
    "Arn": "arn:aws:iam::795974654805:user/Thar@0108"
}
tharunraj@LAPTOP-P0PRBVKB:~$ cd project-1
tharunraj@LAPTOP-P0PRBVKB:~/project-1$ cd App
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App$ ls
Dockerfile  README.md  deployment.yml  dist  service.yml
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App$ cd dist
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App/dist$ ls
assets  index.html  vite.svg
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App/dist$ cd ..
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App$ aws ecr create-repository \
> --repository-name brain-tasks-app \
> --region ap-south-1
{
    "repository": {
        "repositoryArn": "arn:aws:ecr:ap-south-1:795974654805:repository/brain-tasks-app",
        "registryId": "795974654805",
        "repositoryName": "brain-tasks-app",
        "repositoryUri": "795974654805.dkr.ecr.ap-south-1.amazonaws.com/brain-tasks-app",
        "createdAt": "2026-02-10T10:40:52.802000+05:30",
        "imageTagMutability": "MUTABLE",
        "imageScanningConfiguration": {
            "scanOnPush": false
        },
        "encryptionConfiguration": {
            "encryptionType": "AES256"
        }
    }
}
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App$ aws ecr get-login-password --region ap-south-1 \
> docker login --username aws --password-stdin \
> 795974654805.dkr.ecr.ap-south-1.amazonaws.com/brain-tasks-app

Unknown options: docker,login,--username,aws,--password-stdin,795974654805.dkr.ecr.ap-south-1.amazonaws.com/brain-tasks-app
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App$ aws ecr get-login-password --region ap-south-1 | docker login --username AWS --password-stdin 7959746
54805.dkr.ecr.ap-south-1.amazonaws.com

WARNING! Your credentials are stored unencrypted in '/home/tharunraj/.docker/config.json'.
Configure a credential helper to remove this warning. See
```

```
        "repositoryUri": "795974654805.dkr.ecr.ap-south-1.amazonaws.com/brain-tasks-app",
        "createdAt": "2026-02-10T10:40:52.802000+05:30",
        "imageTagMutability": "MUTABLE",
        "imageScanningConfiguration": {
            "scanOnPush": false
        },
        "encryptionConfiguration": {
            "encryptionType": "AES256"
        }
    }
}
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App$ aws ecr get-login-password --region ap-south-1 \
> docker login --username aws --password-stdin \
> 795974654805.dkr.ecr.ap-south-1.amazonaws.com/brain-tasks-app

Unknown options: docker,login,--username,aws,--password-stdin,795974654805.dkr.ecr.ap-south-1.amazonaws.com/brain-tasks-app
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App$ aws ecr get-login-password --region ap-south-1 | docker login --username AWS --password-stdin 7959746
54805.dkr.ecr.ap-south-1.amazonaws.com

WARNING! Your credentials are stored unencrypted in '/home/tharunraj/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App$ docker images
REPOSITORY                  TAG           IMAGE ID       CREATED        SIZE
brain-tasks                 latest        dec8493ebb77   5 days ago     62.2MB
thra21/guvi-project-repo    brain-tasks   dec8493ebb77   5 days ago     62.2MB
<none>                      <none>        fd968376a7aa   10 days ago    64.5MB
<none>                      <none>        88566152beca   10 days ago    64.5MB
trend-web                   latest        54a278583450   10 days ago    71.1MB
thra21/guvi-project-repo    trend-web     54a278583450   10 days ago    71.1MB
devops-build_app            latest        9a7a5c7d5967   10 days ago    64.5MB
nginx                       alpine        2a855eac5070   13 days ago    61.9MB
thra21/guvi-docker-task     1.3           59e0ae90ca43   2 months ago   153MB
thra21/guvi-docker-task     1.2           4c7e99e9925b   2 months ago   75.3MB
thra21/guvi-docker-task     latest        8efbcf199d04   2 months ago   152MB
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App$ docker tag brain-tasks:latest 795974654805.dkr.ecr.ap-south-1.amazonaws.com/brain-tasks:latest
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App$
```

```
REPOSITORY                  TAG           IMAGE ID       CREATED        SIZE
brain-tasks                 latest        dec8493ebb77   5 days ago     62.2MB
thra21/guvi-project-repo    brain-tasks   dec8493ebb77   5 days ago     62.2MB
<none>                      <none>        fd968376a7aa   10 days ago    64.5MB
<none>                      <none>        88566152beca   10 days ago    64.5MB
trend-web                   latest        54a278583450   10 days ago    71.1MB
thra21/guvi-project-repo    trend-web     54a278583450   10 days ago    71.1MB
devops-build_app            latest        9a7a5c7d5967   10 days ago    64.5MB
nginx                       alpine        2a855eac5070   13 days ago    61.9MB
thra21/guvi-docker-task     1.3           59e0ae90ca43   2 months ago   153MB
thra21/guvi-docker-task     1.2           4c7e99e9925b   2 months ago   75.3MB
thra21/guvi-docker-task     latest        8efbcf199d04   2 months ago   152MB
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App$ docker tag brain-tasks:latest 795974654805.dkr.ecr.ap-south-1.amazonaws.com/brain-tasks:latest
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App$ docker push 795974654805.dkr.ecr.ap-south-1.amazonaws.com/brain-tasks:latest
The push refers to repository [795974654805.dkr.ecr.ap-south-1.amazonaws.com/brain-tasks]
92f3917e298a: Preparing
8b4b77877dd1: Preparing
06d5972670c6: Preparing
e22da3f459ae: Preparing
536851695bd2: Preparing
14b08f05bc0a: Waiting
2f3eeb9b695d: Waiting
ae33d91d45a6: Waiting
989e799e6349: Waiting
name unknown: The repository with name 'brain-tasks' does not exist in the registry with id '795974654805'
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App$ docker tag brain-tasks:latest 795974654805.dkr.ecr.ap-south-1.amazonaws.com/brain-tasks-app:latest
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App$ docker push 795974654805.dkr.ecr.ap-south-1.amazonaws.com/brain-tasks-app:latest
The push refers to repository [795974654805.dkr.ecr.ap-south-1.amazonaws.com/brain-tasks-app]
92f3917e298a: Pushed
8b4b77877dd1: Pushed
06d5972670c6: Pushed
e22da3f459ae: Pushed
536851695bd2: Pushed
14b08f05bc0a: Pushed
2f3eeb9b695d: Pushed
ae33d91d45a6: Pushed
989e799e6349: Pushed
latest: digest: sha256:f14ade3add6a338fb3e1cb2ed0818d7be01d1ed0b0e8debe9466592683923598 size: 2199
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App$
```


Repositories | Elastic Container
ap-south-1.console.aws.amazon.com/ecr/private-registry/repositories?region=ap-south-1
Amazon ECR > Private registry > Repositories
Amazon Elastic Container Service
Private registry
Repositories
Features & Settings
Public registry
Repositories
Settings
ECR public gallery
Amazon ECS
Amazon EKS
Getting started
Documentation
Private repositories (1)
View push commands
Delete
Actions
Create repository
Search by repository substring
Repository name
URI
Created at
Tag immutability
Encryption type
brain-tasks-app
795974654805.dkr.ecr.ap-south-1.amazonaws.com/brain-tasks-app
February 10, 2026, 10:40:52 (UTC+05.5)
Mutable
AES-256
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

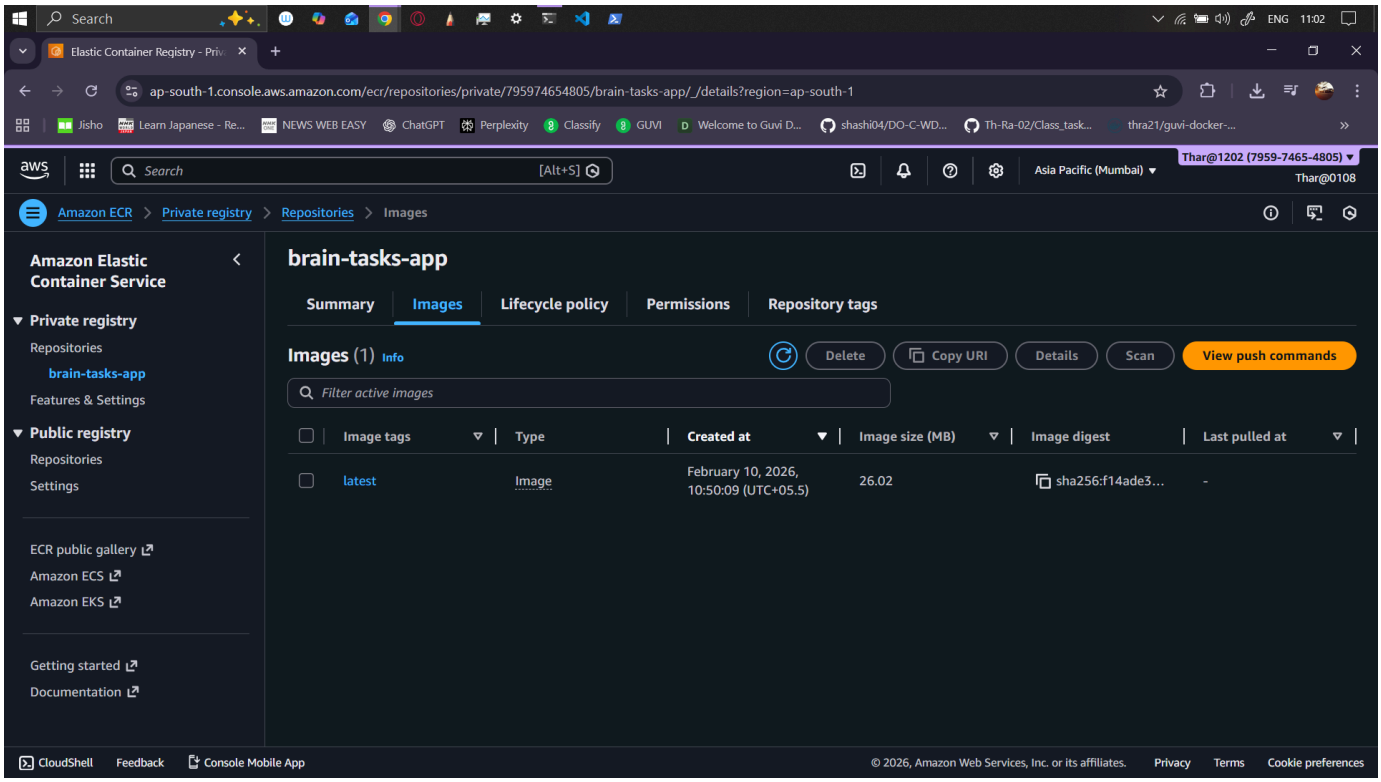

Elastic Container Registry - Priv
ap-south-1.console.aws.amazon.com/ecr/repositories/private/795974654805/brain-tasks-app/_/details?region=ap-south-1
Amazon ECR > Private registry > Repositories > Images
Amazon Elastic Container Service
Private registry
Repositories
brain-tasks-app
Features & Settings
Public registry
Repositories
Settings
ECR public gallery
Amazon ECS
Amazon EKS
Getting started
Documentation
brain-tasks-app
Summary
Images
Lifecycle policy
Permissions
Repository tags
Images (1) Info
Delete
Copy URI
Details
Scan
View push commands
Filter active images
Image tags
Type
Created at
Image size (MB)
Image digest
Last pulled at
latest
Image
February 10, 2026, 10:50:09 (UTC+05.5)
26.02
sha256:f14ade3...
-
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

Fig 1.3-1.7

4.2 Amazon EKS

- An EKS cluster is used to run your application containers in production.
- Within the root repository on your local machine create a folder named “k8s”
- Within the folder create 2 files named “deployment.yml” and “service.yml”

Node Configuration:

- Managed node group
- Two worker nodes (for high availability)
- In this case for the project only one node is used for simplicity.

Kubernetes Resources:

Deployment:

- Manages application pods
- Ensures desired replica count
- Pulls image from ECR

Service:

- Type: LoadBalancer
- Exposes application publicly
- Maps port 80 to container port 3000

Commands used for validation:

- **kubectl deploy k8s/**
- **kubectl get pods**
- **kubectl get svc** (note the external IP)
- **kubectl describe pod <pod-name>**
- **kubectl logs <pod-name>**

```
}
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App$ eksctl create cluster \
  --name brain-eks \
  --region ap-south-1 \
  --nodegroup-name brain-nodes \
  --node-type t3.small \
  --nodes 1 \
  --managed
2026-02-10 11:15:15 [ℹ]  eksctl version 0.220.0
2026-02-10 11:15:15 [ℹ]  using region ap-south-1
2026-02-10 11:15:16 [ℹ]  setting availability zones to [ap-south-1c ap-south-1b ap-south-1a]
2026-02-10 11:15:16 [ℹ]  subnets for ap-south-1c - public:192.168.0.0/19 private:192.168.96.0/19
2026-02-10 11:15:16 [ℹ]  subnets for ap-south-1b - public:192.168.32.0/19 private:192.168.128.0/19
2026-02-10 11:15:16 [ℹ]  subnets for ap-south-1a - public:192.168.64.0/19 private:192.168.160.0/19
2026-02-10 11:15:16 [ℹ]  nodegroup "brain-nodes" will use "" [AmazonLinux2023/1.32]
2026-02-10 11:15:16 [!]  Auto Mode will be enabled by default in an upcoming release of eksctl. This means managed node groups and managed netwo
rking add-ons will no longer be created by default. To maintain current behavior, explicitly set 'autoModeConfig.enabled: false' in your cluster
 configuration. Learn more: https://eksctl.io/usage/auto-mode/
2026-02-10 11:15:16 [ℹ]  using Kubernetes version 1.32
2026-02-10 11:15:16 [ℹ]  creating EKS cluster "brain-eks" in "ap-south-1" region with managed nodes
2026-02-10 11:15:16 [ℹ]  will create 2 separate CloudFormation stacks for cluster itself and the initial managed nodegroup
2026-02-10 11:15:16 [ℹ]  if you encounter any issues, check CloudFormation console or try 'eksctl utils describe-stacks --region=ap-south-1 --cl
uster=brain-eks'
2026-02-10 11:15:16 [ℹ]  Kubernetes API endpoint access will use default of {publicAccess=true, privateAccess=false} for cluster "brain-eks" in
"ap-south-1"
2026-02-10 11:15:16 [ℹ]  CloudWatch logging will not be enabled for cluster "brain-eks" in "ap-south-1"
2026-02-10 11:15:16 [ℹ]  you can enable it with 'eksctl utils update-cluster-logging --enable-types={SPECIFY-YOUR-LOG-TYPES-HERE (e.g. all)} --r
egion=ap-south-1 --cluster=brain-eks'
2026-02-10 11:15:16 [ℹ]  default addons metrics-server, vpc-cni, kube-proxy, coredns were not specified, will install them as EKS addons
2026-02-10 11:15:16 [ℹ]
2 sequential tasks: { create cluster control plane "brain-eks",
    2 sequential sub-tasks: {
        2 sequential sub-tasks: {
            1 task: { create addons },
            wait for control plane to become ready,
        },
        create managed nodegroup "brain-nodes",
    }
}
```

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: brain-tasks-deployment
spec:
  replicas: 1
  selector:
    matchLabels:
      app: brain-tasks-app
  template:
    metadata:
      labels:
        app: brain-tasks-app
    spec:
      containers:
      - name: brain-tasks-app
        image: 795974654805.dkr.ecr.ap-south-1.amazonaws.com/brain-tasks-app:latest
        ports:
        - containerPort: 3000
```

-- INSERT -- 17,84 All

```yaml
apiVersion: v1
kind: Service
metadata:
  name: brain-tasks-service
spec:
  type: LoadBalancer
  selector:
    app: brain-tasks-app
  ports:
    - port: 3000
    targetPort: 80
```

"service.yml" 12L, 177B 8,24 All

```
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App$ aws eks list-clusters --region ap-south-1
{
    "clusters": [
        "brain-eks"
    ]
}
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App$ aws cloudformation list-stacks \
  --stack-status-filter CREATE_FAILED ROLLBACK_COMPLETE ROLLBACK_FAILED CREATE_IN_PROGRESS DELETE_FAILED
{
    "StackSummaries": []
}
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App$ aws eks describe-cluster \
  --name brain-eks \
  --region ap-south-1 \
  --query "cluster.status"
"ACTIVE"
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App$ aws eks update-kubeconfig --region ap-south-1 --name brain-eks
Added new context arn:aws:eks:ap-south-1:795974654805:cluster/brain-eks to /home/tharunraj/.kube/config
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App$ kubectl get nodes
NAME                                          STATUS   ROLES    AGE   VERSION
ip-192-168-36-164.ap-south-1.compute.internal   Ready    <none>   13m   v1.32.9-eks-ecaa3a6
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App$ aws eks describe-cluster --name brain-eks --region ap-south-1 --query "cluster.status"
"ACTIVE"
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App$ ls
Dockerfile  README.md  deployment.yml  dist  service.yml
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App$ vim deployment.yml
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App$ vim service.yml
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App$ docker images
REPOSITORY                                                  TAG           IMAGE ID       CREATED        SIZE
795974654805.dkr.ecr.ap-south-1.amazonaws.com/brain-tasks-app   latest        dec8493ebb77   5 days ago     62.2MB
brain-tasks                                                 latest        dec8493ebb77   5 days ago     62.2MB
thra21/guvi-project-repo                                    brain-tasks   dec8493ebb77   5 days ago     62.2MB
<none>                                                      <none>        fd968376a7aa   10 days ago    64.5MB
<none>                                                      <none>        88566152beca   10 days ago    64.5MB
trend-web                                                   latest        54a278583450   11 days ago    71.1MB
thra21/guvi-project-repo                                    trend-web     54a278583450   11 days ago    71.1MB
devops-build_app                                            latest        9a7a5c7d5967   11 days ago    64.5MB
nginx                                                       alpine        2a855eac5070   13 days ago    61.9MB
thra21/guvi-docker-task                                     1.3           59e0ae90ca43   2 months ago   153MB
```

```
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App$ docker images
REPOSITORY                                                  TAG           IMAGE ID       CREATED        SIZE
795974654805.dkr.ecr.ap-south-1.amazonaws.com/brain-tasks-app   latest        dec8493ebb77   5 days ago     62.2MB
brain-tasks                                                 latest        dec8493ebb77   5 days ago     62.2MB
thra21/guvi-project-repo                                    brain-tasks   dec8493ebb77   5 days ago     62.2MB
<none>                                                      <none>        fd968376a7aa   10 days ago    64.5MB
<none>                                                      <none>        88566152beca   10 days ago    64.5MB
trend-web                                                   latest        54a278583450   11 days ago    71.1MB
thra21/guvi-project-repo                                    trend-web     54a278583450   11 days ago    71.1MB
devops-build_app                                            latest        9a7a5c7d5967   11 days ago    64.5MB
nginx                                                       alpine        2a855eac5070   13 days ago    61.9MB
thra21/guvi-docker-task                                     1.3           59e0ae90ca43   2 months ago   153MB
thra21/guvi-docker-task                                     1.2           4c7e99e9925b   2 months ago   75.3MB
thra21/guvi-docker-task                                     latest        8efbcf199d04   2 months ago   152MB
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App$ vim service.yml
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App$ kubectl apply -f deployment.yml
deployment.apps/brain-tasks-deployment created
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App$ kubectl apply -f service.yml
error: error parsing service.yml: error converting YAML to JSON: yaml: line 10: did not find expected '-' indicator
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App$ ls
Dockerfile  README.md  deployment.yml  dist  service.yml
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App$ vi service.yml
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App$ kubectl apply -f service.yml
error: error parsing service.yml: error converting YAML to JSON: yaml: line 10: did not find expected '-' indicator
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App$ vim deployment.yml
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App$ vim service.yml
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App$ kubectl apply -f service.yml
service/brain-tasks-service created
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App$ kubectl get pods
NAME                                   READY   STATUS    RESTARTS   AGE
brain-tasks-deployment-cf9c74b7-8prps   1/1     Running   0          2m34s
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App$ kubectl get svc
NAME                 TYPE           CLUSTER-IP      EXTERNAL-IP                                                              PORT(S)
AGE
brain-tasks-service  LoadBalancer   10.100.138.82   a602f871169804f4eac12f0fb348d046-588819865.ap-south-1.elb.amazonaws.com   3000:30621/TCP
31s
kubernetes           ClusterIP      10.100.0.1      <none>                                                                   443/TCP
25m
tharunraj@LAPTOP-P0PRBVKB:~/project-1/App$
```

Fig 1.8-1.13

4.3 AWS CodeBuild

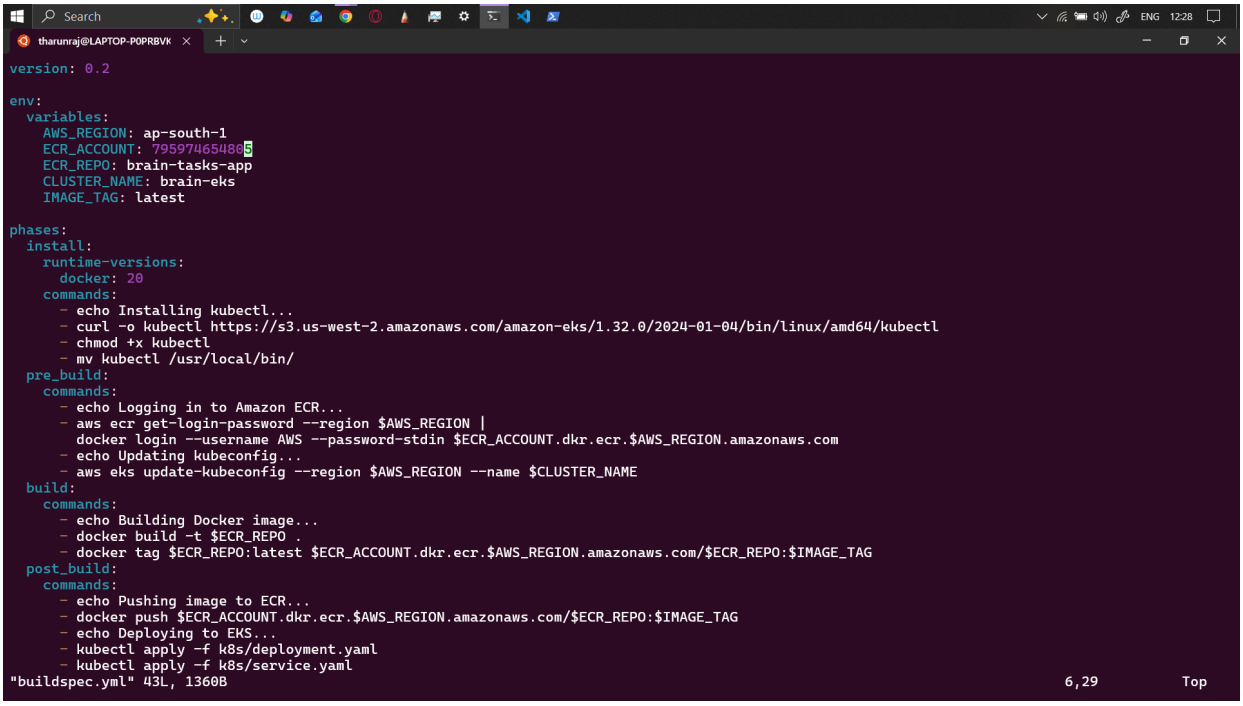
CodeBuild is used to implement the Continuous Integration pipeline.

Responsibilities:

- Authenticate with Amazon ECR
- Build Docker image
- Tag image appropriately
- Push image to ECR
- Update kubeconfig
- Deploy application to EKS using kubectl

Build Phases:

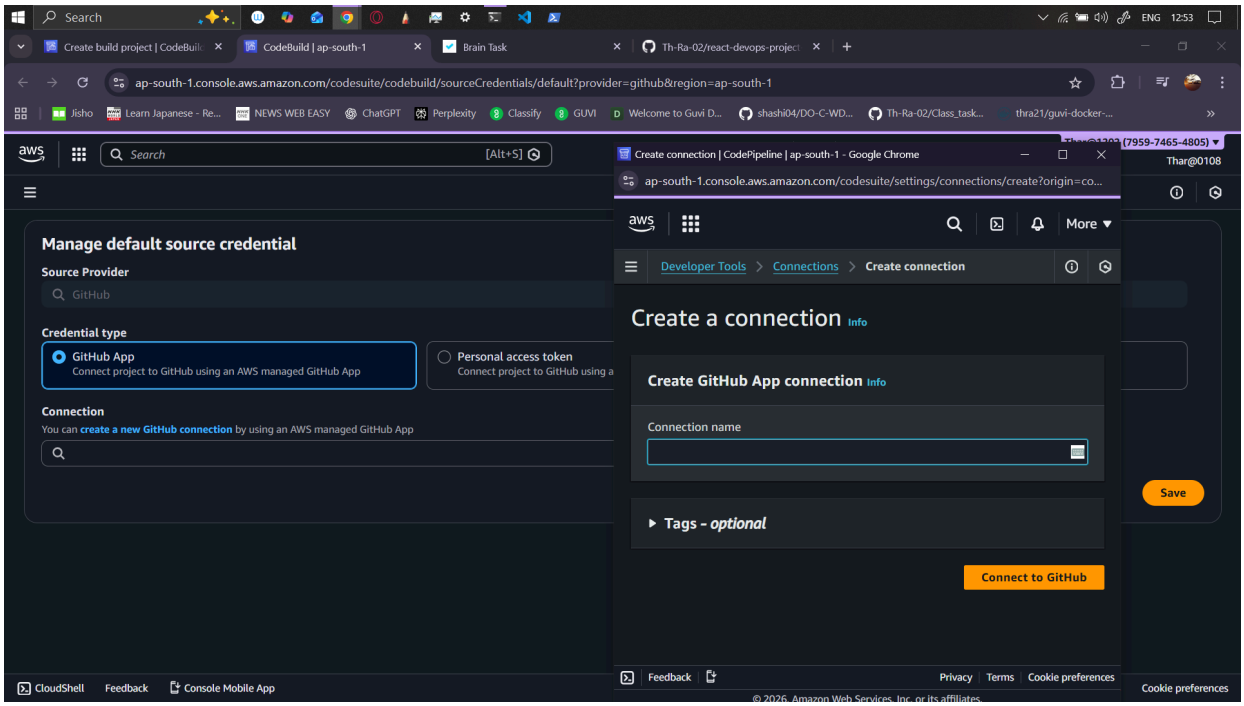
- Create a new file called “**buildspec.yml**”
- Pre-Build Phase Login to ECR using AWS CL
- Build Phase:Docker build command
- Post-Build Phase:Docker push to ECR “**kubectl apply -f k8s/**” to deploy manifests
- Build logs are stored in “**/aws/codebuild/brain-codebuild**”.

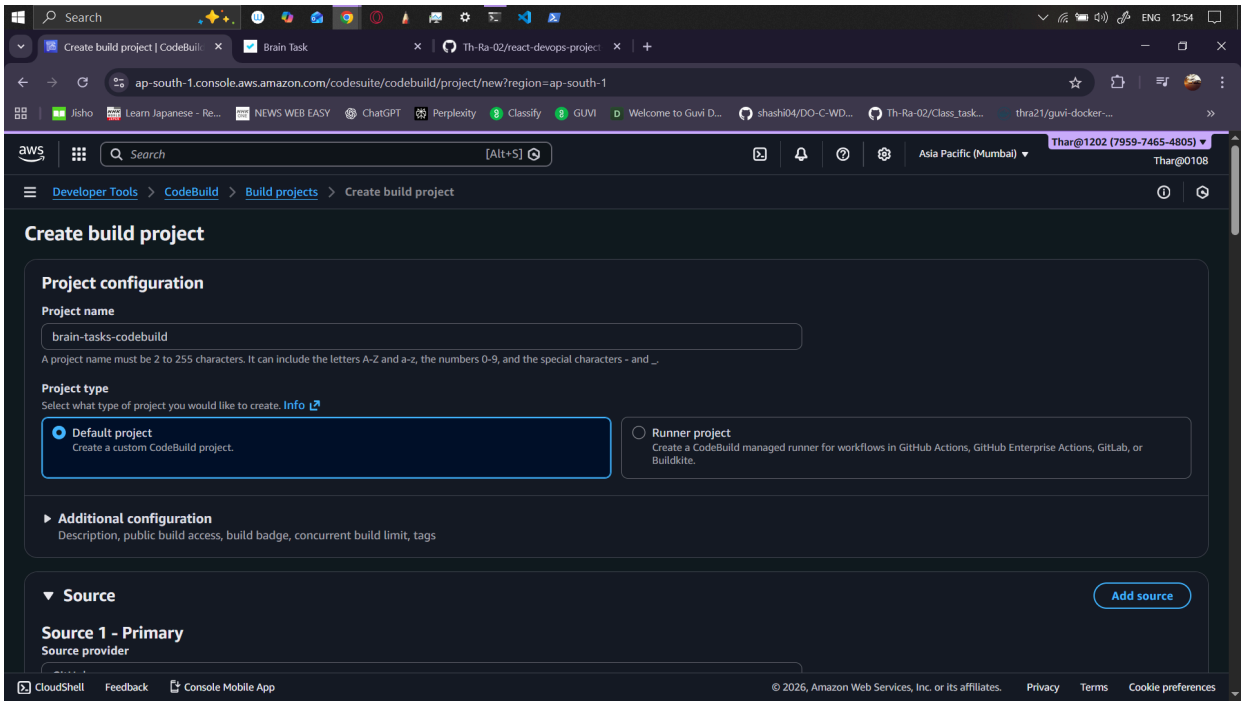


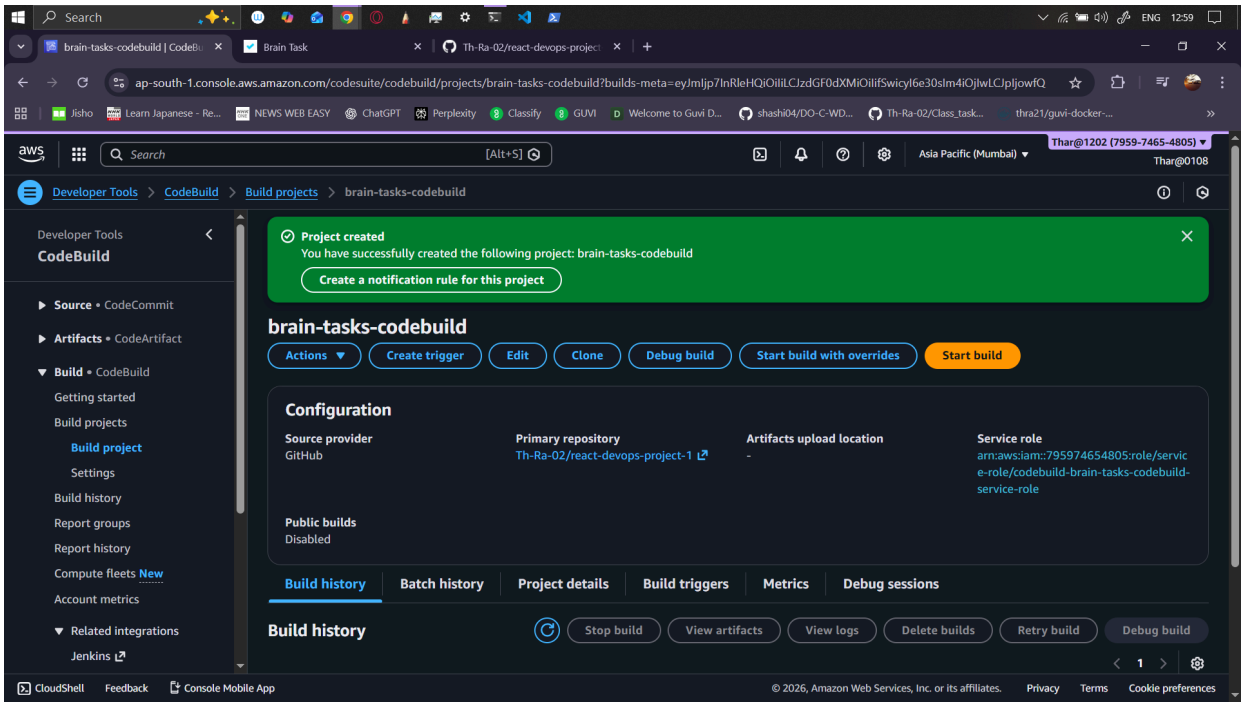
```yaml
version: 0.2

env:
  variables:
    AWS_REGION: ap-south-1
    ECR_ACCOUNT: 795974654805
    ECR_REPO: brain-tasks-app
    CLUSTER_NAME: brain-eks
    IMAGE_TAG: latest

phases:
  install:
    runtime-versions:
      docker: 20
    commands:
      - echo Installing kubectl...
      - curl -o kubectl https://s3.us-west-2.amazonaws.com/amazon-eks/1.32.0/2024-01-04/bin/linux/amd64/kubectl
      - chmod +x kubectl
      - mv kubectl /usr/local/bin/
  pre_build:
    commands:
      - echo Logging in to Amazon ECR...
      - aws ecr get-login-password --region $AWS_REGION |
        docker login --username AWS --password-stdin $ECR_ACCOUNT.dkr.ecr.$AWS_REGION.amazonaws.com
      - echo Updating kubeconfig...
      - aws eks update-kubeconfig --region $AWS_REGION --name $CLUSTER_NAME
  build:
    commands:
      - echo Building Docker image...
      - docker build -t $ECR_REPO .
      - docker tag $ECR_REPO:latest $ECR_ACCOUNT.dkr.ecr.$AWS_REGION.amazonaws.com/$ECR_REPO:$IMAGE_TAG
  post_build:
    commands:
      - echo Pushing image to ECR...
      - docker push $ECR_ACCOUNT.dkr.ecr.$AWS_REGION.amazonaws.com/$ECR_REPO:$IMAGE_TAG
      - echo Deploying to EKS...
      - kubectl apply -f k8s/deployment.yaml
      - kubectl apply -f k8s/service.yaml
"buildspec.yml" 43L, 1360B                                                    6,29          Top
```

```yaml
    ECR_ACCOUNT: 795974654805
    ECR_REPO: brain-tasks-app
    CLUSTER_NAME: brain-eks
    IMAGE_TAG: latest

phases:
  install:
    runtime-versions:
      docker: 20
    commands:
      - echo Installing kubectl...
      - curl -o kubectl https://s3.us-west-2.amazonaws.com/amazon-eks/1.32.0/2024-01-04/bin/linux/amd64/kubectl
      - chmod +x kubectl
      - mv kubectl /usr/local/bin/
  pre_build:
    commands:
      - echo Logging in to Amazon ECR...
      - aws ecr get-login-password --region $AWS_REGION |
        docker login --username AWS --password-stdin $ECR_ACCOUNT.dkr.ecr.$AWS_REGION.amazonaws.com
      - echo Updating kubeconfig...
      - aws eks update-kubeconfig --region $AWS_REGION --name $CLUSTER_NAME
  build:
    commands:
      - echo Building Docker image...
      - docker build -t $ECR_REPO .
      - docker tag $ECR_REPO:latest $ECR_ACCOUNT.dkr.ecr.$AWS_REGION.amazonaws.com/$ECR_REPO:$IMAGE_TAG
  post_build:
    commands:
      - echo Pushing image to ECR...
      - docker push $ECR_ACCOUNT.dkr.ecr.$AWS_REGION.amazonaws.com/$ECR_REPO:$IMAGE_TAG
      - echo Deploying to EKS...
      - kubectl apply -f k8s/deployment.yaml
      - kubectl apply -f k8s/service.yaml
      - kubectl rollout status deployment/brain-tasks-deployment
artifacts:
  files:
    - '**/*'
```

43,0-1 Bot




Create build project | CodeBuild
Brain Task
Th-Ra-02/react-devops-project
ap-south-1.console.aws.amazon.com/codesuite/codebuild/project/new?region=ap-south-1
Asia Pacific (Mumbai)
Thar@1202 (7959-7465-4805)
Thar@0108
Developer Tools > CodeBuild > Build projects > Create build project
Create build project
Project configuration
Project name
brain-tasks-codebuild
A project name must be 2 to 255 characters. It can include the letters A-Z and a-z, the numbers 0-9, and the special characters - and _.
Project type
Select what type of project you would like to create. Info
Default project
Create a custom CodeBuild project.
Runner project
Create a CodeBuild managed runner for workflows in GitHub Actions, GitHub Enterprise Actions, GitLab, or Buildkite.
Additional configuration
Description, public build access, build badge, concurrent build limit, tags
Source
Add source
Source 1 - Primary
Source provider
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


brain-tasks-codebuild | CodeBuild
Brain Task
Th-Ra-02/react-devops-project
ap-south-1.console.aws.amazon.com/codesuite/codebuild/projects/brain-tasks-codebuild?builds-meta=eyJmIjp7InRleHQiOiIiLCJzdGF0dXMiOiIifSwicyI6e30sIm4iOjIwLCJpIjowfQ
Asia Pacific (Mumbai)
Thar@1202 (7959-7465-4805)
Thar@0108
Developer Tools > CodeBuild > Build projects > brain-tasks-codebuild
Developer Tools
CodeBuild
Source • CodeCommit
Artifacts • CodeArtifact
Build • CodeBuild
Getting started
Build projects
Build project
Settings
Build history
Report groups
Report history
Compute fleets New
Account metrics
Related integrations
Jenkins
Project created
You have successfully created the following project: brain-tasks-codebuild
Create a notification rule for this project
brain-tasks-codebuild
Actions
Create trigger
Edit
Clone
Debug build
Start build with overrides
Start build
Configuration
Source provider
GitHub
Primary repository
Th-Ra-02/react-devops-project-1
Artifacts upload location
-
Service role
arn:aws:iam::795974654805:role/service-role/codebuild-brain-tasks-codebuild-service-role
Public builds
Disabled
Build history
Batch history
Project details
Build triggers
Metrics
Debug sessions
Build history
Stop build
View artifacts
View logs
Delete builds
Retry build
Debug build
CloudShell
Feedback
Console Mobile App
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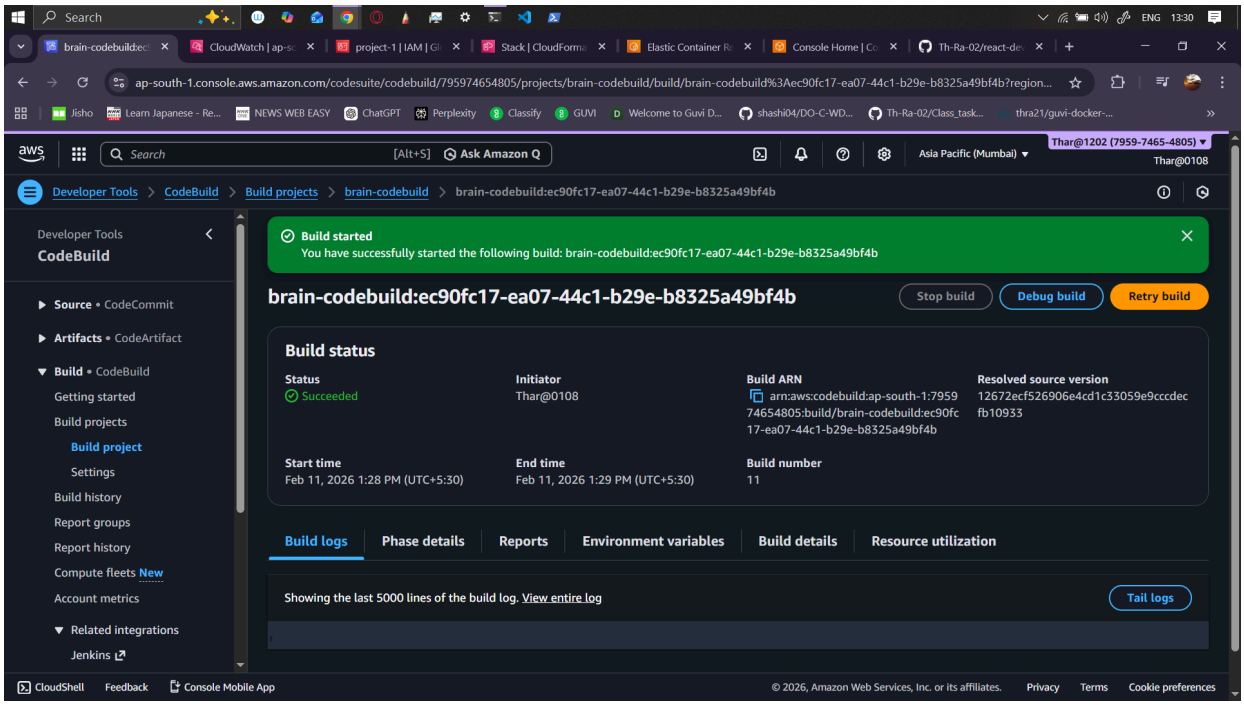
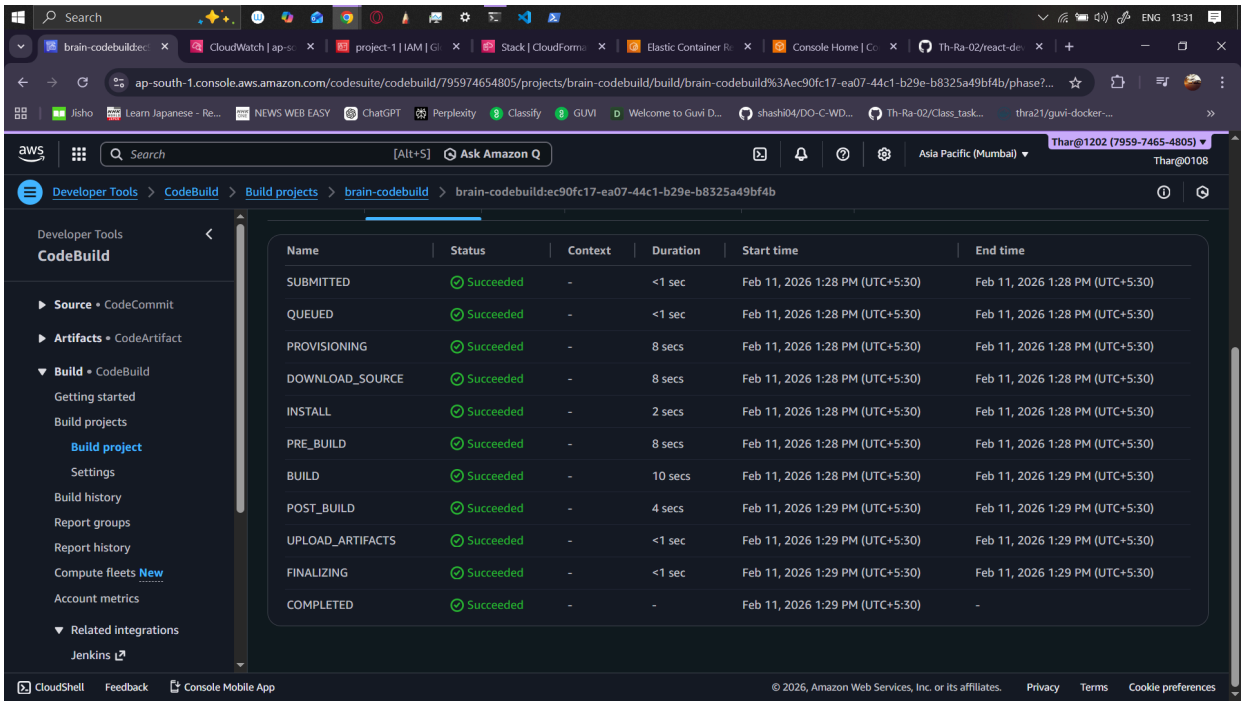

Developer Tools > CodeBuild > Build projects > brain-codebuild > brain-codebuild:ec90fc17-ea07-44c1-b29e-b8325a49bf4b
Build started
You have successfully started the following build: brain-codebuild:ec90fc17-ea07-44c1-b29e-b8325a49bf4b
brain-codebuild:ec90fc17-ea07-44c1-b29e-b8325a49bf4b
Stop build
Debug build
Retry build
Build status
Status
Succeeded
Initiator
Thar@0108
Build ARN
arn:aws:codebuild:ap-south-1:795974654805:build/brain-codebuild:ec90fc17-ea07-44c1-b29e-b8325a49bf4b
Resolved source version
12672ecf526906e4cd1c33059e9cccdecfb10933
Start time
Feb 11, 2026 1:28 PM (UTC+5:30)
End time
Feb 11, 2026 1:29 PM (UTC+5:30)
Build number
11
Build logs
Phase details
Reports
Environment variables
Build details
Resource utilization
Showing the last 5000 lines of the build log. View entire log
Tail logs


Developer Tools > CodeBuild > Build projects > brain-codebuild > brain-codebuild:ec90fc17-ea07-44c1-b29e-b8325a49bf4b
Developer Tools
CodeBuild
Source • CodeCommit
Artifacts • CodeArtifact
Build • CodeBuild
Getting started
Build projects
Build project
Settings
Build history
Report groups
Report history
Compute fleets New
Account metrics
Related integrations
Jenkins
Name | Status | Context | Duration | Start time | End time
SUBMITTED | Succeeded | - | <1 sec | Feb 11, 2026 1:28 PM (UTC+5:30) | Feb 11, 2026 1:28 PM (UTC+5:30)
QUEUED | Succeeded | - | <1 sec | Feb 11, 2026 1:28 PM (UTC+5:30) | Feb 11, 2026 1:28 PM (UTC+5:30)
PROVISIONING | Succeeded | - | 8 secs | Feb 11, 2026 1:28 PM (UTC+5:30) | Feb 11, 2026 1:28 PM (UTC+5:30)
DOWNLOAD_SOURCE | Succeeded | - | 8 secs | Feb 11, 2026 1:28 PM (UTC+5:30) | Feb 11, 2026 1:28 PM (UTC+5:30)
INSTALL | Succeeded | - | 2 secs | Feb 11, 2026 1:28 PM (UTC+5:30) | Feb 11, 2026 1:28 PM (UTC+5:30)
PRE_BUILD | Succeeded | - | 8 secs | Feb 11, 2026 1:28 PM (UTC+5:30) | Feb 11, 2026 1:28 PM (UTC+5:30)
BUILD | Succeeded | - | 10 secs | Feb 11, 2026 1:28 PM (UTC+5:30) | Feb 11, 2026 1:29 PM (UTC+5:30)
POST_BUILD | Succeeded | - | 4 secs | Feb 11, 2026 1:29 PM (UTC+5:30) | Feb 11, 2026 1:29 PM (UTC+5:30)
UPLOAD_ARTIFACTS | Succeeded | - | <1 sec | Feb 11, 2026 1:29 PM (UTC+5:30) | Feb 11, 2026 1:29 PM (UTC+5:30)
FINALIZING | Succeeded | - | <1 sec | Feb 11, 2026 1:29 PM (UTC+5:30) | Feb 11, 2026 1:29 PM (UTC+5:30)
COMPLETED | Succeeded | - | - | Feb 11, 2026 1:29 PM (UTC+5:30) | -
CloudShell
Feedback
Console Mobile App
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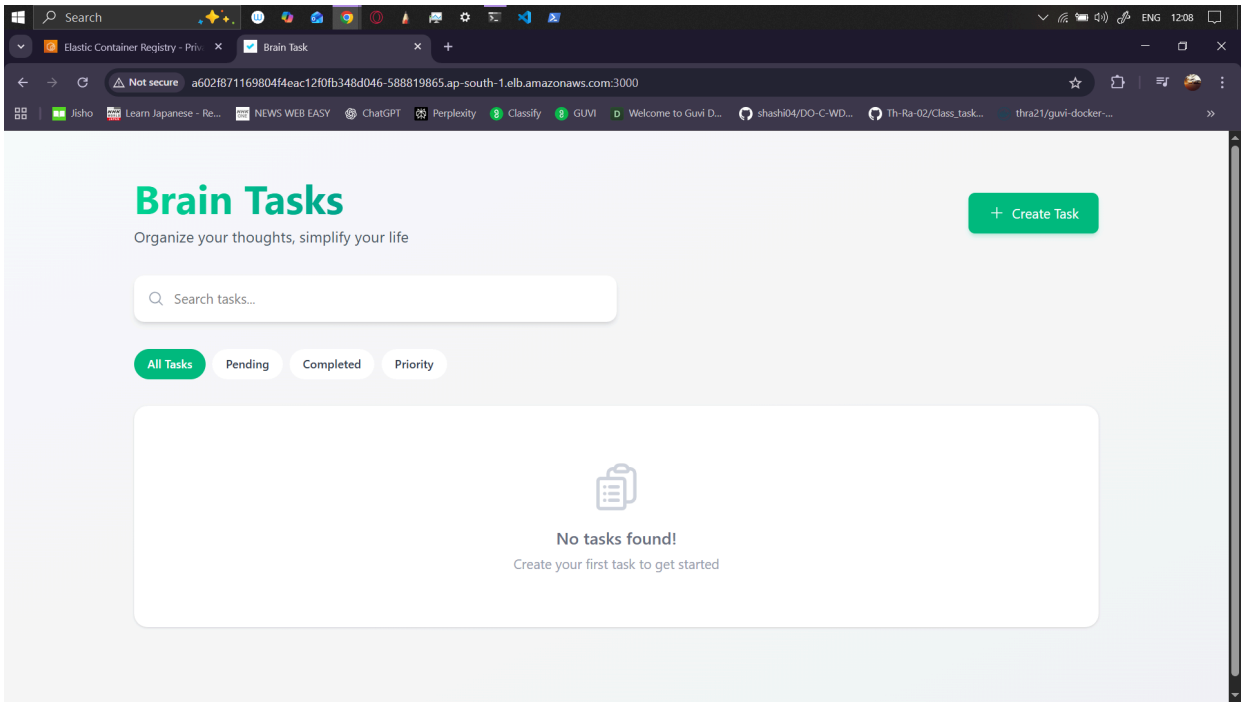
Brain Tasks
Organize your thoughts, simplify your life
+ Create Task
Search tasks...
All Tasks
Pending
Completed
Priority
No tasks found!
Create your first task to get started

Fig 1.14-1.21

4.4 CloudWatch Monitoring and Logging

CloudWatch was configured to provide three levels of observability.

CodeBuild Logs - Log Group: /aws/codebuild/brain-codebuild
EKS Cluster Metrics (Container Insights) - Log Group: /aws/containerinsights/brain-eks/performance
Application Logs (Fluent Bit Integration)
Log Group: /aws/containerinsights/brain-eks/application

Setup Included:

- IAM OIDC provider configuration
- IAM service account for CloudWatch
- AWS managed policy (CloudWatchAgentServerPolicy)
- Helm installation of:
  - aws-cloudwatch-metrics
  - aws-for-fluent-bit

This enables centralized monitoring of:

- Pod logs
- Cluster performance
- Resource utilization
- Application output logs

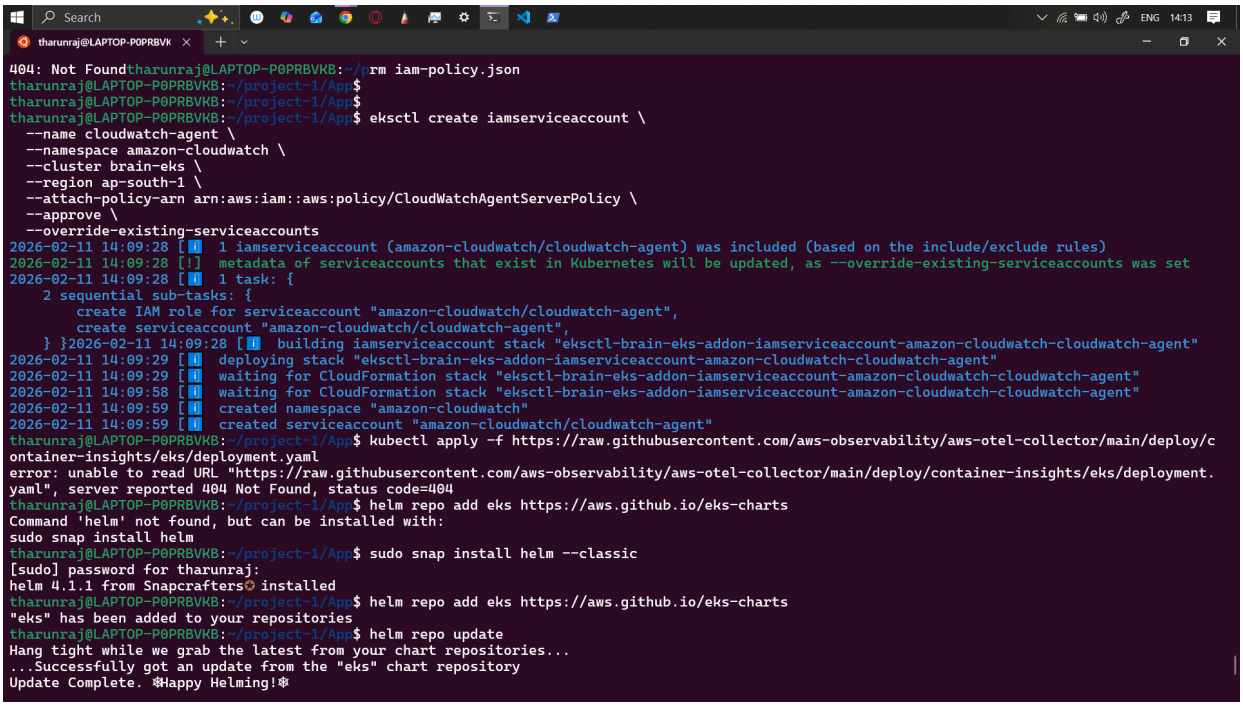

Fig 1.22 and 1.23

5. CI/CD Workflow

- Code Push: Developer commits and pushes changes.
- Build Triggered: CodeBuild starts automatically.
- StImage Creation: Docker image is built from Dockerfile.
- Image Push: Image is pushed to Amazon ECR.
- Deployment: kubectl apply command updates Kubernetes deployment.
- Kubernetes Update: EKS pulls new image and updates pods.
- Exposure: LoadBalancer service exposes application publicly.
- Monitoring: CloudWatch captures logs and performance metrics.

6. Security Configuration

Security measures implemented include:

- IAM roles for CodeBuild
- OIDC provider for EKS
- IAM service account for CloudWatch agent
- AWS managed policies
- Secure ECR authentication
- Kubernetes RBAC integration

This ensures controlled access and secure deployment.

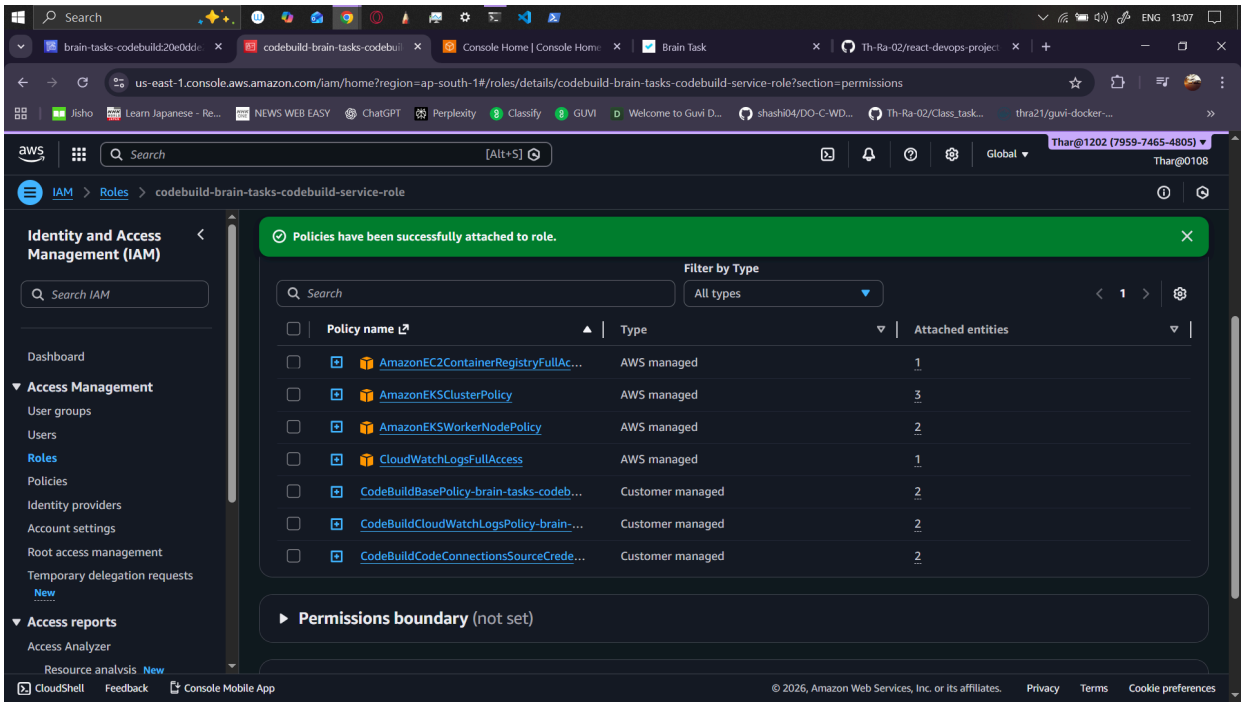

Fig 1.24

7. Validation and Testing

The system was validated through the following checks:

1. Successful CodeBuild execution.
2. Docker image availability in ECR.
3. Running pods in EKS.
4. External LoadBalancer endpoint created.
5. Application accessible via browser.
6. Build logs visible in CloudWatch.
7. Application logs visible in CloudWatch.
8. Cluster performance metrics available.

8. Final Outcome

The project successfully implements a production-style CI/CD pipeline using AWS managed services.

Key achievements:

- Automated container build process
- Image registry integration
- Kubernetes-based scalable deployment
- Public application exposure
- Centralized logging
- Cluster monitoring and observability
- Secure IAM-based access control

9. Conclusion

This project successfully implements a complete CI/CD pipeline using AWS services to build, containerize, deploy, and monitor a web application on Amazon EKS. The integration of CodeBuild, ECR, EKS, and CloudWatch demonstrates automated deployment, scalable container orchestration, and centralized logging within a secure cloud environment. Overall, the solution reflects practical DevOps principles and real-world cloud infrastructure implementation.

10. Repository Links

- Github Repository:
- https://github.com/Th-Ra-02/react-devops-project-1
- Dockerhub Repository:
- https://hub.docker.com/repository/docker/thra21/guvi-project-repo/general